

SAKATA, Japan

WMO: 475870

Lat: 38.9086N

Lon: 139.8440E

Elev: 16

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.4	28.0	15.2	11.9	32.2	17.1	13.1	33.0	30.1	34.3	27.8	33.2	13.9	310	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.5	90.1	76.6	87.5	75.9	85.0	75.1	78.3	86.1	77.4	84.7	76.3	83.3	9.6	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.3	137.2	82.5	75.2	132.2	81.8	74.1	127.3	80.9	41.9	86.4	40.9	84.8	39.9	83.6	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.0	23.1	20.7	DB	22.6	95.2	2.6	2.8	20.7	97.3	19.2	98.9	17.7	100.5	15.8	102.5
			WB	21.1	80.1	2.4	1.0	19.4	80.8	17.9	81.4	16.6	82.0	14.8	82.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.6	35.2	35.7	41.0	50.3	60.7	68.2	75.5	78.5	71.2	60.3	49.4	40.1	(6)
	DBStd	15.81	3.91	4.53	5.05	5.93	5.42	4.41	4.50	4.35	5.50	5.10	5.77	4.87	(7)
	HDD50	1599	460	400	283	66	1	0	0	0	0	1	80	308	(8)
	HDD65	4512	925	820	745	440	154	19	0	0	8	161	469	771	(9)
	CDD50	3656	0	0	4	77	333	547	791	883	637	321	61	2	(10)
	CDD65	1094	0	0	0	1	21	116	326	418	196	16	0	0	(11)
	CDH74	7654	0	0	0	6	114	421	2211	3767	1104	31	0	0	(12)
	CDH80	2176	0	0	0	0	11	57	607	1239	259	3	0	0	(13)
(14) Wind	WSAvg	9.5	12.5	11.6	10.7	9.6	8.5	7.6	7.1	7.4	8.0	8.8	10.2	12.1	(14)
(15) Precipitation	PrecAvg	75.7	7.1	4.6	4.4	4.0	4.7	5.2	8.5	7.3	6.3	6.9	8.2	8.5	(15)
	PrecMax	91.0	11.0	8.3	7.4	6.2	10.0	13.7	25.4	14.5	13.3	11.1	15.5	12.9	(16)
	PrecMin	54.0	4.2	1.9	2.4	0.8	1.6	1.2	3.2	1.7	3.0	3.5	3.1	3.1	(17)
	PrecStd	8.5	1.7	1.5	1.3	1.4	2.2	3.1	4.4	3.6	2.7	1.9	2.7	2.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.3	53.5	63.0	74.0	81.4	85.4	92.3	94.5	90.6	77.4	68.0	57.6	(19)
		MCWB	44.9	45.5	49.8	56.8	63.1	69.8	77.1	77.4	75.8	66.3	58.5	51.7	(20)
	2%	DB	45.4	48.4	56.9	68.5	77.1	81.2	88.8	91.2	86.1	73.7	64.4	53.0	(21)
		MCWB	40.8	43.2	47.5	55.3	61.7	68.0	75.8	77.2	74.6	64.0	56.6	48.1	(22)
	5%	DB	43.1	45.3	52.8	64.1	73.8	78.5	85.9	88.7	82.7	71.5	61.2	50.4	(23)
		MCWB	38.7	40.3	45.9	52.5	60.4	67.2	75.4	76.8	73.1	62.8	54.8	45.5	(24)
	10%	DB	41.3	42.8	49.7	60.4	70.6	76.1	83.3	86.4	79.9	69.1	58.6	47.8	(25)
		MCWB	36.8	37.8	44.1	51.4	59.7	66.4	74.3	76.1	71.6	61.4	52.9	43.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.5	47.7	51.8	60.3	66.4	73.1	78.9	80.1	77.7	69.3	61.3	53.0	(27)
		MCDB	48.4	50.9	58.3	66.8	74.2	78.8	87.9	87.7	86.5	74.3	65.2	56.2	(28)
	2%	WB	41.7	43.8	49.0	57.0	64.3	71.1	77.4	79.0	75.7	66.7	58.3	48.7	(29)
		MCDB	44.5	47.7	54.5	65.2	72.2	76.4	85.1	86.7	83.4	71.2	62.4	52.4	(30)
	5%	WB	39.4	41.0	46.9	54.7	62.8	69.7	76.4	78.1	74.4	64.5	55.8	46.0	(31)
		MCDB	42.4	45.0	51.8	61.2	70.2	75.2	83.5	85.5	81.2	69.8	60.0	49.9	(32)
	10%	WB	37.3	38.4	44.7	52.8	61.2	68.3	75.3	77.2	72.9	62.6	53.5	43.3	(33)
		MCDB	40.6	42.3	49.8	59.0	68.5	74.0	81.9	84.4	78.9	67.8	58.3	47.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	9.5	12.0	14.6	14.2	12.1	11.2	12.5	13.0	13.7	12.4	9.4	(35)
		MCDBR	11.4	14.4	19.5	21.7	20.7	16.7	15.1	15.5	15.4	16.2	16.7	13.7	(36)
	5% WB	MCWBR	9.1	11.0	12.9	11.8	9.7	7.5	6.5	6.4	7.2	9.0	11.6	11.3	(37)
		MCWBR	11.6	14.1	16.5	16.7	15.6	12.2	13.0	13.2	13.5	13.0	14.7	13.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.408	0.443	0.509	0.546	0.556	0.540	0.551	0.523	0.463	0.435	0.426	0.417	(40)
		taud	1.959	1.871	1.813	1.805	1.844	1.969	2.005	2.092	2.218	2.174	2.098	2.005	(41)
	Edn at Noon		226	236	235	236	237	241	237	240	248	240	221	212	(42)
		Edh at Noon	42	53	62	66	65	57	55	49	41	39	37	38	(43)
(44) All-Sky Solar Radiation	RadAvg		427	678	1046	1436	1612	1636	1442	1502	1212	908	563	377	(44)
	RadStd		37	71	96	142	168	140	194	174	89	69	51	41	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.56	N/A	N/A	N/A	N/A

Nomenclature: See separate page